basic education
Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

2023/24 ANNUAL TEACHING PLANS: TECHNICAL MATHEMATICS: GRADE 10 (TERM 1)

TERM 1	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11
DATE COMPLETED											
CAPS TOPICS	**NUMBER SYSTEMS (INCLUDING BINARY NUMBERS AND INTRODUCTION OF COMPLEX NUMBERS)**			**EXPONENTS**		**ALGEBRAIC EXPRESSIONS**					**REVISION/ASSESSMENT**
TOPICS/CONCEPTS, SKILLS AND VALUES	1. Identify rational numbers and convert terminating or recurring decimals into the form $\frac{a}{b}$ where $a,b\in \mathbf{Z}$ and $b \neq 0$. 2. Introduce binary and complex numbers 3. Determine between which two integers a given simple surd lies 4. Round real numbers to an appropriate degree of accuracy (to a given number of decimal digits)			1. Simplify expressions using the laws of exponents for integral exponents 2. Solve exponential equations (accepting that the laws of exponents hold for real exponents and solutions are not necessarily integral or even rational) 3. Revise scientific notation		1. Revise notation (interval, set builder, number line, sets) 2. Adding and subtracting of algebraic terms 3. Multiplication of a binomial by a binomial 4. Multiplication of a binomial by a trinomial 5. Determine the HCF and LCM of not more than three numerical or monomial algebraic expressions by making use of factorisation 6. Factorise: • Common factor (revision) • Grouping in pairs • Difference of two squares (revision) • Trinomials • Difference and sums of two cubes; and 7. Algebraic fractions • Simplifying, adding, subtracting, multiplying and division with numerators and denominators limited to the polynomials covered in factorisation			All Term 1 topics
PAT & SBA	**ASSIGNMENT & TEST**										

2023/24 ANNUAL TEACHING PLANS: TECHNICAL MATHEMATICS: GRADE 10 (TERM 2)

TERM 2	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11
DATE COMPLETED											

CAPS TOPICS	**EQUATIONS AND INEQUALITIES**	**TRIGONOMETRY**	**REVISION/ASSESSMENT**
TOPICS/CONCEPTS, SKILLS AND VALUES	1. Revise notation (interval, set builder, number line, sets) 2. Solve linear equations 3. Solve equation with fractions 4. Manipulation of formulae (change subject of the formula) 5. Solve quadratic equations by factorisation of the form: • $x^2 + bx + c = 0$ • $-x^2 + bx + c = 0$ $ax^2 + bx + c = 0$ where a is a common factor 6. Solve simultaneous linear equations with two variables 7. Solve word problems involving linear, quadratic, or simultaneous linear equations 8. Solve simple linear inequalities	1. Know definitions of the trigonometric ratios sin θ, cos θ and tan θ, using right-angled triangles for 0°≤ θ ≤ 360° 2. Introduce the reciprocals of the 3 basic trigonometric ratios, sin θ, cos θ and tan θ 3. Trigonometric ratios in each of the quadrants are calculated where one ratio in the quadrant is given by making use of diagrams (No variables) 4. Practise the use of a calculator for questions applicable to trigonometry 5. Solve simple trigonometric equations for angles between 0° and 90° 6. Solve two-dimensional problems involving right angled triangles (no variables) 7. Trigonometry graphs: • $y = asin\theta, y = acos\theta$ and $y = atan\theta$ for $0° \leq \theta \leq 360°$ • $y = asin\theta + q$ and $y = acos\theta + q$ for $0° \leq \theta \leq 360°$	Term 1 & 2 content
PAT & SBA	**PAT 1 & TEST /JUNE EXAMIINATION**		

2023/24 ANNUAL TEACHING PLANS: TECHNICAL MATHEMATICS: GRADE 10 (TERM 3)

TERM 3	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11
DATE COMPLETED											
CAPS TOPICS	**EUCLIDEAN GEOMETRY**					**ANALYTICAL GEOMETRY**		**FUNCTIONS AND GRAPHS**			**REVISION/ASSESSMENT**
TOPICS/CONCEPTS, SKILLS AND VALUES	1.Revise basic geometry done in Grades 8 and 9 • Lines and parallel lines, angles, triangles congruency and similarity • Pythagoras' theorem • Calculate the unknown side of a right-angled triangle. 2. Apply the properties of line segments joining the mid-points of two sides of a triangle. Do practical problems. 3. Know the features of the following special quadrilaterals: the kite, parallelogram, rectangle, rhombus, square and 4. Trapezium (apply to practical problems)					1. Represent geometric figures on a Cartesian co-ordinate system 2. Apply for any two points $(x_1; y_1)$ $(x_2; y_2)$ formulae for determining the: • Distance between the two points • Gradient of the line segment connecting the two points (and from that identify parallel and perpendicular lines) • Coordinates of the mid-point of the line segment joining the two points; and • The equation of a straight line passing through two points. $y = mx + c$		1. Functional notation 2. Generate graphs by means of point-by-point plotting supported by available technology. 3. Drawing of the following functions: • Linear function • Quadratic function • Hyperbola • Exponential			Term 3 content
PAT & SBA	**PAT 2 & TEST**										

2023/24 ANNUAL TEACHING PLANS: TECHNICAL MATHEMATICS: GRADE 10 (TERM 4)

<table>
<tr><th>TERM 4</th><th>WEEK 1</th><th>WEEK 2</th><th>WEEK 3</th><th>WEEK 4</th><th>WEEK 5</th><th>WEEK 6</th><th>WEEK 7</th><th>WEEK 8</th><th>WEEK 9</th><th>WEEK 10</th></tr>
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<tr><td>DATE COMPLETED</td><td></td><td></td><td></td><td></td><td colspan="6"></td></tr>
<tr><td>CAPS TOPICS</td><td>MENSURATION</td><td>CIRCLES, ANGLES, AND ANGULAR MOVEMENT</td><td colspan="2">FINANCE AND GROWTH</td><td colspan="6">REVISION & EXAMS</td></tr>
<tr><td>TOPICS/CONCEPTS, SKILLS AND VALUES</td><td>Conversion of units, square units, and cubic units.</td><td>1. Define a radian
2. Indicate the relationship between degrees and radians, convert radians to degrees or degrees to radians and minutes to radians and radians to degrees and minutes</td><td colspan="2">1. Use the simple and compound growth formulae and to solve problems, including interest hire purchase, inflation, population growth and other real-life problems
2. Understanding the implication of fluctuation foreign exchange rates (e.g., on the petrol price, imports, exports, overseas travel)</td><td colspan="6">All topics/concepts, skills and values</td></tr>
<tr><td>SBA</td><td colspan="10">FINAL EXAMINATION (PAPER 1 & PAPER 2)</td></tr>
<tr><td>EXAMINATION</td><td colspan="10">
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<tr><th colspan="2">PAPER 1</th><th colspan="2">PAPER 2</th></tr>
<tr><th>TOPIC</th><th>MARKS</th><th>TOPIC</th><th>MARKS</th></tr>
<tr><td>Algebra (Number system, exponents, expressions, equations and inequalities))</td><td>60 ± 3</td><td>Analytical geometry</td><td>15 ± 3</td></tr>
<tr><td>Functions & graphs</td><td>25 ± 3</td><td>Trigonometry</td><td>40 ± 3</td></tr>
<tr><td>Finance, growth, and decay</td><td>15 ± 3</td><td>Euclidean geometry</td><td>30 ± 3</td></tr>
<tr><td></td><td></td><td>Mensuration and circles, angles, and angular movement</td><td>15 ± 3</td></tr>
<tr><td>TOTAL</td><td>100</td><td>TOTAL</td><td>100</td></tr>
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